Model of the Mouse Visual-Motor System

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Abstract

The goal of this project is to design a conceptual deep network model of the mouse visuo-motor system as it relates to predator threat detection and defensive behaviour. The model spans two major brain systems. First is the visual system, from the retina through the SC and dLGN to the primary visual cortex, and secondly, the defensive motor system, from the SC's divergent subcortical projections through the PAG down to spinal motor output. We chose this particular behaviour, freeze vs. escape in response to aerial threats, because it is one of the few cases where the full input-to-output circuit is known in enough detail to meaningfully constrain a deep network design. The model is trained in a simulated environment with two stimulus classes: looming (a rapidly expanding overhead shadow approximating a hawk diving towards the mouse) and sweeping (non-threatening background motion). Outputs are either a freeze vector or a directed escape velocity vector. This setup necessitated a dual-pathway architecture, not because it was the obvious engineering choice, but as a reflection of biological constraints. The fast subcortical pathway handles the low-latency reflexive case; the cortical pathway handles context. The intended application is a fundamental understanding of sensorimotor integration: what circuit structure is actually necessary to go from raw visual input to survival-appropriate motor output, and does a network trained to solve that problem naturally mirror the functional organization of the mouse brain?

1 Background

The mouse (Mus musculus) has become one of the preeminent model organisms in systems neuroscience, largely because its brain is compact enough to study at circuit-level resolution while still facilitating an extensive range of naturalistic behaviours. One behaviour in particular has attracted sustained attention: the defensive response to overhead visual threats. When a mouse detects the rapid radial expansion of an overhead dark mass (i.e, a diving hawk), it will either freeze in place or bolt toward the nearest shelter, depending on a host of factors. These two responses are not only different in degree; they are mechanistically distinct programs driven by different downstream circuits [13]. Freeze is generally favoured when the threat is ambiguous or distant, as staying still conserves energy and reduces detection [10][27]. Escape dominates when the stimulus is imminent, and shelter is within reach. From a modelling perspective, this system is particularly compelling due to the high degree of characterization of its underlying circuitry; the key structures, their axonal projections, and the mutually inhibitory logic of behavioural selection are now resolved with a level of precision rare within systems neuroscience.

Visual processing begins at the retina, where distinct classes of retinal ganglion cells encode motion and contrast and send signals downstream along two routes that broadly correspond to fast reflexive processing and slower cortical evaluation. The retino-collicular pathway projects directly to the superficial superior colliculus (sSC), a midbrain structure with a topographic retinotopic map that can trigger rapid defensive responses without involving the cortex at all, looming-evoked freezing and escape survive lesions of V1, confirming the SC can operate largely autonomously [46][24]. In parallel, the retino-geniculate-cortical pathway relays signals through the dorsal lateral geniculate nucleus (dLGN) of the thalamus to the primary visual cortex and higher visual areas, supporting a more deliberate evaluation of what the threat actually is and where it is going [45]. These two pathways are not fully independent, in which the cortex feeds back onto the SC through corticotectal projections, but for the purposes of modelling threat responses, the retino-collicular route is the faster and more critical one.

At the motor interface, the deep superior colliculus (dSC) functions as the primary hub for behavioral selection. Parvalbumin-positive (PV+) excitatory neurons in the dSC project to the lateral posterior (LP) thalamus, which relays through the central amygdala (CeA) to drive the ventrolateral periaqueductal gray (vlPAG) and produce freezing [37][45]. A parallel dSC projection targets the parabigeminal nucleus (PBG), activating the basolateral amygdala (BLA) and the dorsal PAG (dPAG) to drive high-speed escape [37][13]. Because the vlPAG and dPAG are reciprocally inhibitory, they form a competitive gating circuit that determines the final motor output [41]. This escape drive ultimately reaches the mesencephalic locomotor region (MLR), activating spinal central pattern generators (CPGs) to produce the stepping movements needed for flight [4]. Finally, secondary motor cortex (M2) provides top-down modulation to both PAG branches, allowing factors such as knowledge of shelter location to bias the behavioral selection process [11].

Rather than reconstructing circuits from granular single-neuron biophysics, the goal-driven paradigm optimizes networks for functional behavioural objectives to uncover the normative computational logic underlying biological systems [26]. The mouse defensive system is a robust candidate for this approach, as it combines a

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well-defined neural blueprint with ethologically valid stimuli that are computationally simple to render and simulate.

2 Model Design

2.1 Visual System

The visual stimuli that trigger defensive behaviours in mice are highly parameter-dependent, and understanding these features, despite the low visual acuity of mice, directly motivates our choice of model inputs [39]. The key insight from early behavioural work is that the defensive response is not a general reaction to visual change, but rather a highly selective response to a specific configuration of features.

The visual display of a looming dark disk triggers rapid escape or freezing in mice, and these reactions are strongly dependent on the parameters of the visual stimulus. Specifically, the stimulus must be a dark disk on a bright background, rapidly expanding and present overhead, and altering the parameters drastically reduces the behavioural response [46]. The mouse does not respond to expanding white disks, dimming dark disks, or contracting white disks. All of these stimuli share some low-level features with the expanding dark disk, but not the overall configuration [22]. This selectively implies that the visual system is not simply detecting contrast change or motion energy in general, but is performing a specific computation over the temporal expansion profile of a dark object in the upper visual field.

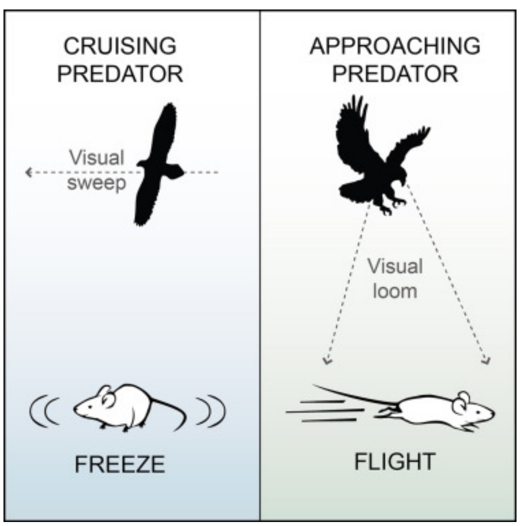

Figure 1: Graphical abstraction of mouse freeze and flight defensive behaviours based on overhead threat observation[10].

The opposing behavioural response, freezing, is triggered by a qualitatively different stimulus class. A small moving disk, stimulating the sweep of a predator cruising overhead, induces freezing in mice, meaning that looming and sweeping stimuli provide visual triggers for opposing flight and freeze behaviours [10]. These two stimulus classes, the rapidly expanding looming disk and the small laterally sweeping disk, provide biologically grounded and experimentally validated inputs that map directly into our two training stimulus classes.

Critical information for threat classification isn't present in any single frame, but rather in the temporal evolution of the stimulus, specifically whether the object is expanding, at what rate, and from which direction. Hence, the input to the visual module is a temporal sequence of frames rather than a single image. Deep SC neurons, unlike superficial SC neurons, respond selectively to the looming feature with spatial invariance and habituation to familiar stimuli, none of which are actually present in the retinal signal outlined in the next section. This indicates that these properties emerge from processing across time and space within the SC itself [22]. With this biological grounding, we represent the model input as a stack of consecutive frames, which gives the retinal and SC modules access to the temporal gradient necessary to compute expansion rate. The retinal component is analogous to how transient On-Off direction-selective retinal ganglion cells encode motion by comparing responses across short temporal windows, allowing the network to learn the expansion signature that distinguishes looming aerial predator signals from stationary or receding stimuli.

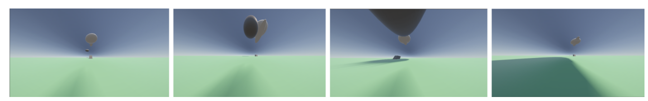

Figure 2: Time-series progression of the looming visual stimulus from the agent's egocentric perspective. The sequence illustrates the rapid radial expansion of a low-luminance spherical mass, biologically simulating an imminent predator dive. While depicted here in RGB color for visual clarity, this continuous sequence is fed into the retino-collicular pathway as a low-resolution grayscale tensor.

2.2 Visual Information Processing

The mouse visual system is organized through local circuits within specific neural structures and long range connections that link these areas to transmit spatial and feature based information. Retinotopy and eye specificity are the two most common ways scientists map visual space [35]. Figure 3 shows the mapping space from the retina to the brain in the mouse. Neighboring points in a visual scene are mapped onto neighboring neurons in the retina. Eye specificity refers to how axonal projections from the two eyes segregate into distinct domains within brain targets that receive binocular input. This eye specificity is visualized by injecting different colored anterograde tracers into each eye. dLGN neurons send their axons to V1, which in mice is composed of two zones. The monocular zone (M) receives information via dLGN exclusively from the contralateral eye, whereas the binocular zone (B) gets input from both eyes [35].

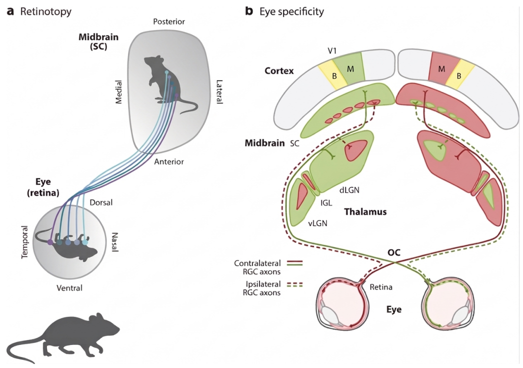

Figure 3: Retinotopy and eye specificity in the mouse visual system. Neighboring retinal locations project to neighboring target regions, while monocular and binocular domains segregate across the visual pathway.

The Retinal Foundation vision begins at the photoreceptor mosaic, which places a constraint on downstream items. The mouse retina is rod-dominated, specialized for low-light scotopic conditions. The retinal photoreceptor mosaic generates neural signals that represent the image reaching the eye. Approximately 3% of mouse photoreceptors are cones, where the cones express one or a combination of two photopigments: one sensitive to short wavelengths of light, and one sensitive to medium wavelengths of light. The mouse's visual system is optimized for motion detection and threat avoidance rather than high resolution detail. Since the mouse does not have a fovea, its entire retina is similar to the peripheral retina of a primate [18].

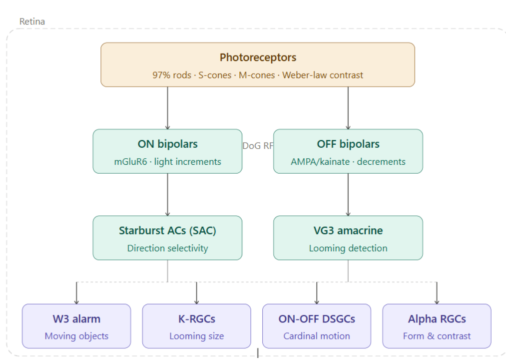

Figure 4: Retinal circuit organization from photoreceptors through ON and OFF bipolar pathways, amacrine cells, and retinal ganglion cell output.

Once photoreceptors turn light into electrical signals, three types of retinal interneurons (horizontal, bipolar, and amacrine cells) process and refine those signals. These cells then pass the information to the retinal ganglion (RGCs), which serve as the output neurons of the eye. The spiking activities of the RGCs are then sent to the brain, where they drive visual percepts and light-mediated behaviors. There are 30+ different RGC types, each of which responds best to a particular aspect of the visual scene. These cells calculate information such as motion direction, edge orientation, and looming threats before the signal leaves the eye [35].

2.3 Vision Architecture

Modelling the mouse vision requires a computational architecture that mirrors the biological specialization of their visual system, particularly for survivor behaviours like predator avoidance. The architecture spans from retinal preprocessing to rapid midbrain driven motor decisions. Figure 4 outlines the vision architecture of mice where it explains the retinal processing prior to motor decisions.

Prior to this spatial filtering, the model initiates with light transduction at the photoreceptor layer. The mouse retina is approximately 97% rod-dominated, with two cone types (S-cones and M-cones) forming an ethologically optimized gradient for UV-rich sky and green-lit ground [14]. This stage implements contrast sensitivity using the Naka–Rushton equation: $R = R_{\max} \cdot \frac{I^n}{I^n + I_{\text{half}}^n}$ where I_{half} is the semi-saturation constant. Retinal light adaptation dynamically shifts the effective operating point with background luminance, approximating Weber's law ($\Delta I / I = \text{const}$).[29][31].

After photoreceptors, signals split into ON and OFF pathways at the bipolar cell layer. ON bipolars express mGluR6 (metabotropic), inverting the sign of photoreceptor output so they depolarise to light increments. OFF bipolars express AMPA/kainate (ionotropic) receptors and depolarise to light decrements [40]. Each bipolar cell's spatial RF is modelled as a Difference of Gaussians, effectively functioning as the biological equivalent of a convolutional kernel in a CNN. The same model applies to RGC centre-surround RFs: $D_s(x, y) = \pm[(1/2\pi\sigma_{cen}^2) \cdot \exp(-(x^2 + y^2)/2\sigma_{cen}^2) - B/(2\pi\sigma_{sur}^2) \cdot \exp(-(x^2+y^2)/2\sigma_{sur}^2)]$ where B controls the balance between center and surround contributions [9].

Amacrine cells in the retina serve as essential interneurons, with Starburst Amacrine Cells (SACs) mediating direction selectivity via spatiotemporal correlation and VG3 cells specializing in looming detection through approach-sensitive glutamatergic signaling [25][20]. These computations inform the 30+ distinct Retinal Ganglion Cell (RGC) types in, which function as parallel feature channels that encode all visual information for subcortical targets. Significant RGC types include the W3 "alarm" neurons, which use nonlinear center pooling to detect small predators against stationary backgrounds, and OFF-transient alpha RGCs (Kcnip2 molecular marker), which encode the size of looming threats to drive immediate defensive responses [47][44]. This parallel architecture also includes ON-OFF Direction-Selective RGCs (DSGCs) for tracking cardinal motion and sustained alpha-like cells that relay form and contrast details to the primary visual cortex [25].

The visual processing described above, from the photoreceptor transduction to retinal interneurons, and the parallel output channels of the RGCs, produce a set of feature-selective signals that are transmitted downstream along two anatomically distinct routes, each of which feeds a different part of the motor system.

The first and faster route is the retino-collicular pathway, in which RGC axons project directly to the superficial superior colliculus (sSC). This pathway bypasses the cortex and forms the biological basis of the fast subcortical motor pathway described in the next section. The SC receives input from approximately 85-90% of mouse RGCs, making it the dominant retinal target for the species, and is where the retinal feature channels described above converge to drive the premotor decision process [35].

The second and slower route is the retino-geniculate-cortical pathway, in which a subset of RGC axons relay through the dorsal lateral geniculate nucleus (dLGN) of the thalamus to the primary visual cortex (V1) and higher visual areas. This pathway supports a more deliberate evaluation of the visual threat, providing the contextual inputs that modulate defensive thresholds and bias escape direction, and is detailed in the motor system architecture.

2.4 Motor System - Fast Pathway

The first step of the fast pathway is implemented as a single convolutional layer that receives a RGC feature tensor from the retinal module and models the spatial integration of RGC inputs onto sSC neurons, capturing direction and looming selectivity across the retinotopic map. The shallow depth is deliberate, as fewer layers means shorter processing time, reflecting the small number of synapses in the retino-collicular route. The SC is kept cortex-free as the fast reflexive pathway operates largely independently of cortex, as stated in Background. The interlaminar sSC to dSC projection is a feedforward pathway with no cortical gating, represented in the model as a linear projection with ReLU and LayerNorm. The sSC also encodes a topographic heading signal from the most active region of the retinotopic map, and this scalar is propagated through the escape branch and combined with the cuneiform urgency signal at the final escape projection layer to produce a directed velocity output.

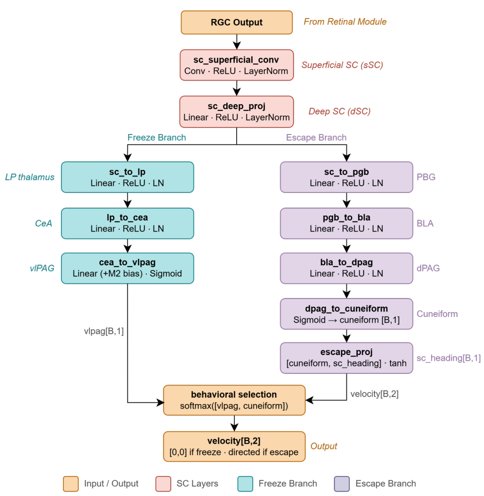

Figure 5: Fast subcortical pathway diagram linking retinal output to the superior colliculus and downstream freeze and escape branches.

2.4.1 Freeze Branch. The dSC parvalbumin+ (PV+) excitatory neurons project to the LP thalamus, which relays the threat signal to the CeA, as mentioned in Background. The CeA activates vlPAG output neurons via disinhibition (GABAergic CeA projections release vlPAG neurons from tonic inhibitory suppression), which produces an active locomotion arrest motor program in which reticulospinal projections maintain muscle tone, breathing shallows, and heart rate drops [37].

In the model, three sequential layers, representing SC to LP thalamus, LP thalamus to CeA, and CeA to vlPAG, each implement a linear transform with ReLU and LayerNorm, producing representations of LP Thalamus and CeA populations. The final sigmoid produces a vlPAG scalar that takes on a value between and including 0 and 1, where 1 represents full locomotion arrest. M2 cortical output from the slow pathway also adds a bias term before the final sigmoid, modelling the known M2 to PAG projection that can raise or lower the threshold for freezing based on behavioral context.

2.4.2 Escape Branch. As stated in Background, the dSC projects to the PBG, which drives the BLA. The BLA activates dPAG to evoke wild running or backward fleeing in mice [42], functionally opposed to the CeA's freeze. The dPAG excites the cuneiform nucleus, a key part of the MLR, which acts as the speed-setting input to the CPGs. These stages are handled by the output module.

In the model, four layers, representing SC to PBG, PBG to BLA, BLA to dPAG, and dPAG to cuneiform, follow the same linear-ReLU-LayerNorm pattern, producing population representations at each stage. The dPAG-to-cuneiform layer applies sigmoid to yield a cuneiform scalar encoding locomotor urgency. A final escape projection layer takes the cuneiform speed signal and SC heading and produces a Tanh-bounded velocity vector, reflecting the bidirectional nature of locomotor commands.

2.4.3 Behavioral Selection. The reciprocally inhibitory relationship between vlPAG and dPAG is enforced in the model as a softmax over the vlPAG scalar and the cuneiform scalar, producing a pair of probabilities for each sample. The argmax is recorded as the behavior label, with 0 representing freeze and 1 representing escape. When freeze wins, velocity gets set to zero. The bias from M2 shifts this competition, where a positive bias favours escape and a negative bias favours freeze.

Furthermore, the corticotectal pathway introduces a second, slower input to this interface. Corticotectal inputs from V1 boost the gain of looming-evoked SC responses in a retinotopic manner, reducing SC responses by almost half when optogenetically silenced in awake mice, while leaving looming speed tuning unchanged [48]. In the model, this is implemented as the V1 module's output being concatenated with the SC deep features before the dual output heads, providing a modulatory signal that scales overall threat sensitivity without altering feature selectivity. The V1 gain signal modulates the threshold at which either output stream activates, providing a substrate for context-dependent adjustment of defensive sensitivity. The m2_pag_bias term, which enters the system downstream, represents a further top-down modulatory signal from secondary motor cortex that operates on the motor side of this interface rather than the visual side.

2.5 Motor System - Slow Pathway

2.5.1 Slow Cortical Pathway. This section details the slow cortical pathway and how it results in motor output generation. In Section 2 we discussed how the SC can be modeled to produce subcortical reflex actions. Our slow cortical pathway models how visual stimuli are processed through the cortical hierarchy, which includes the visual system (V1 + HVAs), the motor system (MOp + MOs), and finally translated into coordinated movement by spinal circuits (MLR). To ground the scale of our model, we note the following biological constraints on the mouse brain:

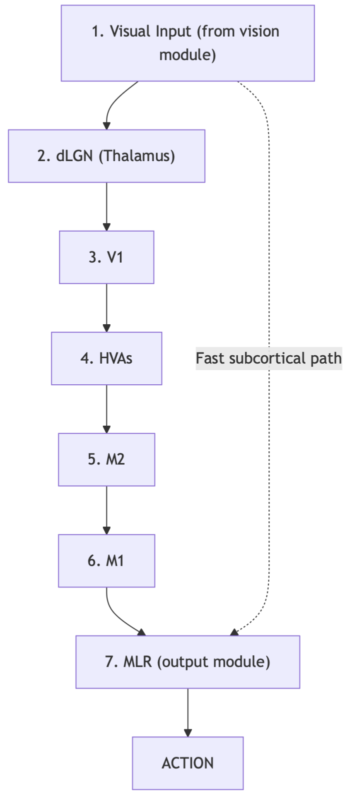

Figure 6: Slow cortical pathway from visual input through dLGN, V1, HVAs, M2, M1, and MLR to action output.

- Total cortical volume is approximately 124 mm^3
- On average density of ~80K excitatory neurons per mm^3
- The sensory cortex contains roughly 5 million neurons
- Out of which the visual cortex accounts for about 1 million
- V1/VISp alone has ~230,000 neurons [3]
- V1 is connected to 9 higher visual areas (HVAs)
- dLGN contains ~20,000 thalamocortical projection neurons [12]
- Motor cortex (24 mm^3) = MOs/M2 (13) + MOp/M1 (11)
- MOs (13×80K, ~1.04M) MOp (11×80K, ~880K neurons)

These values serve as constraints on the relative capacity of each module in our model (channel counts, hidden state dimensionality, compression ratios).

2.5.2 Dorsal Lateral Geniculate Nucleus (dLGN). The dLGN in the thalamus serves as the primary relay between retina and cortex. Around 30–40% of synapses onto thalamocortical (TC) relay cells are modulatory inputs from cortex, brainstem, and the thalamic reticular nucleus (TRN), while retinal inputs are in the minority [16]. We model the dLGN as a 2D convolutional layer (1⟶16 channels, 7×7 kernel) with L1 regularization on the kernel weights. While MouseNet [38] used a 9×9 kernel based on ~9° receptive fields from Piscopo et al. (2013), we use a 7×7 kernel to be consistent with more recent findings from Tschetter et al. (2018), who reported adult mouse dLGN receptive field radii stabilizing around 5.3° ± 0.4° (~1 pixel/degree resolution). The L1 penalty on kernel weights enforces that only a small subset of weights dominate each receptive field, approximating the sparse but strong retinal convergence [32].

2.5.3 Visual Cortex - V1 + Higher Visual Areas (HVAs). V1 (VISp) performs the first major cortical transformation using canonical interlaminar (L4 ⟶ L2/3 ⟶ L5) circuitry to extract features and regulate cortical gain via divisive normalization [7, 17]. These signals fan out in parallel to nine higher visual areas (HVAs) with functional specializations shown by Marshel et al. (2011): motion-biased (VISl, VISal, VISrl, VISam), pattern-biased (VISpm), and bimodal (VISli); VISpl, VISa and VISpor are without assigned roles.

Rather than stacked feedforward CNNs (as in MouseNet), we adopt a modified CORnet-S [21], whose within-area recurrence drives brain-likeness on Brain-Score. We introduce CORnet-M (CORM), a mouse-adapted variant that replaces the serial primate chain with parallel branching after VISp: one dedicated V1 block followed by five parallel recurrent blocks for functionally grouped HVAs + VISpl + VISa + VISpor. Each block retains CORnet-S's recurrence, skip connections, batch normalization, and ReLU nonlinearity. All outputs converge at VISpor before feeding the motor pathway. We omit L6 corticothalamic feedback for simplification; L5 corticotectal projections to SC are detailed previously. Full block specifications are provided in Appendix A.

2.5.4 Secondary Motor Cortex - M2/MOs. The secondary motor cortex (MOs/M2) functions as the main planning and goal-selection hub, integrating abstract visual features from the HVAs and maintaining persistent activity that encodes future action goals [34].

We model M2 using a 2-layer GRU with a 128-dimensional hidden state. GRUs capture the recurrent architecture of persistent activity [19, 23] necessary to process a looming stimulus unfolding across multiple video frames and produce continuous (time-involving) outputs. The GRU receives two inputs: (1) from CORM (after VISpor global average pooling), and (2) output from the fast pathway [34]. In this way subcortical threat information is incorporated into cortical planning. The hidden state compresses cortical inputs (HVAs ~10^5–10^6 neurons) to a lower-dimensional decision space (compression ratio ~780:1 to ~7800:1). This is consistent with the M2's role in dimensionality reduction observed between cortical population activity and motor output [15, 34].

2.5.5 Primary Motor Cortex - MOp/M1. The primary motor cortex is the final cortical stage that transforms the abstract plan from M2 into a low-dimensional motor command [34]. In the mouse motor cortex, the 6 principal components capture 97–99% of the variance during dexterous forelimb movement [34], meaning that while many neurons participate, the output that actually drives movement is substantially lower-dimensional than the cortical state.

We model M1 as a linear readout layer that projects the 128-dimensional M2 hidden state onto a 6-d vector. A separate readout layer with Tanh activation then outputs a velocity vector (v_x, v_y). Note that mouse M1 exhibits recurrent structure and population dynamics during movement [34]; however, we simplified the architecture to focus on kinematic planning (temporal pattern generation is delegated to spinal CPGs).

2.6 Output Module

2.6.1 Mesencephalic Locomotor Region (MLR) + Spinal Circuits. The output module integrates input from both slower and faster pathways. MLR serves as the final common pathway for coordinated locomotion [4, 33]. Cortical motor commands project to MLR via subcortical relays that initiate and modulate locomotor rhythms before activating CPGs. We model the MLR as a thresholded gating layer (initiating locomotion only once input exceeds a certain activation level) that translates velocity vectors into movement, including stepping speed and steering [6, 8].

3 Training Plan and Environment

To properly train the dual-pathway visual-motor architecture, the model requires an environment that simulates the physical and optical realities of a mouse evading predation. Rather than relying on static, pre-labelled datasets, this model utilizes a programmatic 3D arena built within the Unity game engine. This allows for continuous tracking of absolute coordinates for the mouse, predator, and shelter. Crucially, because a real mouse cannot learn predator evasion through trial and error, this iterative training process does not represent a single organism learning within its lifetime. Instead, the optimization loop simulates natural selection on an evolutionary timescale, shaping synaptic weights over generations to hardwire optimal survival reflexes into the subcortical fast pathway.

To evaluate the computational viability of this architecture, this environment serves as a Proof of Concept (PoC) demonstrating that Unity ML-Agents can be strictly biologically constrained to study embodied cognition. Unlike standard grid-world models where outputs directly shift an agent's spatial coordinates, our PoC forces the network to solve a true physical control problem. The network's final output vector acts as the drive signal from the Mesencephalic Locomotor Region (MLR) to simulated spinal Central Pattern Generators (CPGs). By mapping these signals to continuous physical forces applied to the agent's mass, the model is subjected to realistic biomechanical penalties: overcoming inertia, limited maximum acceleration, and physical turning radii. This ensures the cortical planning modules (M2) and subcortical reflex arcs are forced to cope with the physical limitations of skeletal muscle during predator evasion.

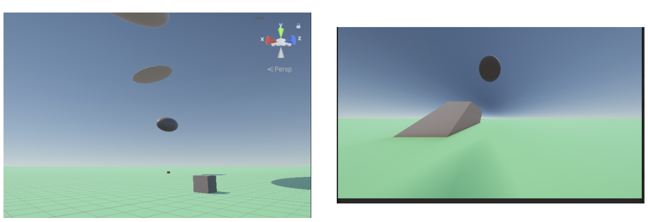

Figure 7: Overhead (left) and prey's-eye (right) perspectives of the simulated visuo-motor environment. Left: The 3D continuous arena. The artificial agent (center) navigates to a physical shelter (grey wedge) to evade predation. Overhead stimuli are divided into a looming predator (low-luminance spherical mass) and harmless sweeping distractors (high-luminance stretched ovals) to train visual selectivity. Right: The raw visual input tensor for the retino-collicular pathway. The 22° upward-pitched camera with an expanded FOV mimics a mouse's binocular overlap. Rendered in RGB but processed as a low-resolution grayscale tensor, the setup forces convolutional layers to classify threats using spatiotemporal luminance rather than textural cues.

Within this environment, the model receives two distinct streams of continuous data to approximate the sensory and spatial awareness of the mouse. The agent's primary sensor is a simulated camera mounted on its physical body in the Unity environment. Anatomically, mice are lateral-eyed, providing each eye with an expansive ~180° monocular visual field and a modest 40°-50° binocular overlap, which creates a near-panoramic field of vision where the binocular field crucially extends directly above the animal's head [1]. To biologically constrain our model to this specific spatial orientation and massive, non-foveated visual space, the Unity camera utilizes a horizontally stretched aspect ratio with an artificially expanded Field of View (FOV).

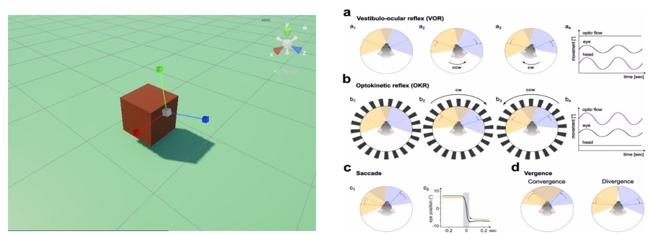

Figure 8: Spatial and visual constraints of the simulated agent. Left: The agent's physical body represented in the Unity environment, illustrating the camera mounting pitched upwards at 22° to capture overhead stimuli. Right: Top-down view of the biological mouse visual field. The blue region indicates the binocular overlap zone (~40°-50°) extending directly above the head; the yellow and pale regions indicate the monocular fields (~180° each). Adapted from Giovanetti & Rancz (2024).

To enforce the rod-dominated biology of scotopic vision, the model's visual input is bottlenecked into a low-resolution grayscale tensor [18]. By stripping away photorealistic textures, the convolutional layers are mathematically forced to rely on temporal expansion and luminance cues to differentiate expanding dark edges from harmless clouds or simple full-field dimming, perfectly mirroring the subsecond threat-detection mechanisms of the biological retino-collicular pathway [46]. Second, an egocentric spatial vector provides the agent's absolute distance and coordinates relative to the shelter. This approximates allocentric mapping [36] and retrosplenial shelter-vector processing [5], enabling the M2 pathway to compute escape trajectories even when visually occluded [2].

3.1 Reinforcement Learning Objective and Loss Function

The network uses Deep Reinforcement Learning (DRL) to map visuo-spatial inputs to survival actions (freeze vs. flight), optimizing synaptic weights to minimize a comprehensive loss function.

Each term translates a biological pressure into a mathematical constraint, naturally incentivizing the emergence of dual behavioural outputs. The model captures the core neuroecological tension of resource gathering versus predation risk by continuously rewarding the agent for exploring the open environment via the Foraging term, while penalizing it with a Death penalty if it collides with the predator [27]. Because sprinting is metabolically exhausting, overriding the baseline CPG rhythm to output high-velocity escape drives incurs a continuous Energy penalty (W2), directly tying physical motor exertion to the loss function [13]. If the shelter is nearby, outputting an escape vector ensures survival, but if caught too far out in the open, escaping guarantees death mid-sprint. Therefore, the model learns to output a freeze vector in exposed areas, incurring zero energy penalty while lowering the probability of death by exploiting the predator's motion-based vision. Fleeing from harmless clouds triggers a False Alarm penalty to enforce strict visual selectivity. Rather than attempting the computationally prohibitive task of modelling exact synaptic delays at the single-neuron level, Reaction Frames simply counts the frames elapsed between the threat's appearance and the agent's first movement, acting as a functional abstraction to enforce the subsecond reflexes of the retino-collicular pathway [46].

3.2 Evaluation Metrics

Success is quantified by the model's ability to dynamically select defensive behaviours based on spatial context. The primary metric is a behavioural bifurcation plot charting action selection (freeze vs. flight) against shelter distance at threat onset. A biologically accurate model should demonstrate a sharp spatial decision boundary: near 100% escape initiation within a survivable radius, transitioning strictly to freezing when exposed [43]. Furthermore, circuit necessity is evaluated via *in silico* lesion studies comparing survival curves across three variants: a "wild-type" dual-pathway model, a "cortically lesioned" model (lacking the spatial vector), and a "collicularly lesioned" model (blinded to the camera). By plotting the survival rates of these isolated pathways against the intact architecture, we can demonstrate why survival requires the integration of both subcortical visual reflexes and cortical spatial planning.

4 Discussion and Future Work

Several biological and computational simplifications made in this design present clear avenues for future iteration. First, to address the sparsity of the escape reward and bridge the gap between gradient descent and biological one-shot learning, future models could implement Neural Episodic Control (NEC) as proposed by Pritzel et al. [30]. Inspired by the hypothesized role of the hippocampus in decision making, NEC utilizes a semi-tabular representation of experience to rapidly assimilate and act upon new experiences. Acting as a computational analogue to the mouse hippocampus [36], an NEC module would allow the agent to rapidly cache and recall the spatial coordinates of the shelter after a single successful escape without waiting for slow gradient updates. Secondly, the manually tuned loss function weights (W1 through W5) could be optimized via evolutionary meta-learning to quantitatively reveal which precise ecological pressures most strongly mandate the dual-pathway architecture. Finally, incorporating dynamic postural geometry, such as the $-29°$ downward head pitch observed during natural murine locomotion [28], would more accurately capture how optic flow and overhead looming stimuli dynamically enter the retinal image during active movement.

5 Conclusion

The mouse defensive circuit is a remarkable feat of evolutionary optimization. Within milliseconds of an overhead shadow expanding, a 20-gram organism performs a critical survival calculation. This process occurs well before any conscious assessment takes place, and it spans visual sensing, behavioral choice, and motor activation. The fact that modern systems neuroscience can trace this computation from the initial retinal photon capture to the spinal central pattern generators (CPGs) driving skeletal muscle provides a profound blueprint for artificial visuo-motor systems. This project translates that biological blueprint into a computational hypothesis. The goal is not merely to describe the circuit structurally, but to quantitatively test whether an artificial network, subjected to identical survival pressures, converges on the same architectural solutions favoured by evolution.

The resulting model bridges the retina, superior colliculus, periaqueductal gray, motor cortex, and spinal cord through a dual-pathway design dictated by fundamental biological trade-offs. The low-latency subcortical pathway governs immediate reflexive responses where cortical processing delays would prove fatal; conversely, the parallel cortical pathway resolves high-order spatial strategies beyond the scope of reflexive circuitry. Crucially, the objective function never explicitly prescribes freezing or flight behaviors. Instead, these discrete behaviours emerge naturally from the neuroecological costs of mortality, metabolic expenditure, and temporal latency.

Ultimately, this architecture serves as more than a design exercise; every structural choice represents a falsifiable hypothesis regarding the constraints the mammalian brain must navigate to solve embodied ecological challenges. Through the proposed *in silico* lesion studies, this framework moves beyond static description. By systematically ablating the rapid subcortical pathway or cortical spatial memory and quantifying the resulting survival failures, this model provides a concrete, rigorous mathematical sandbox to validate our fundamental understanding of sensorimotor integration.

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A CORnet-Mouse (CORM) Architecture Details

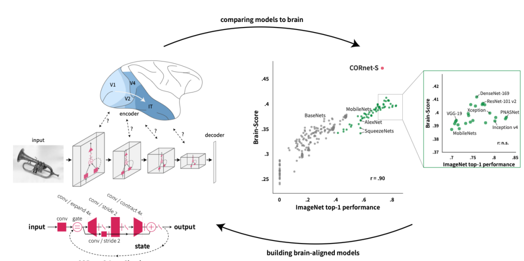

Figure 9: CORnet-S area circuit and benchmark comparison, motivating the mouse-adapted CORnet-Mouse (CORM) architecture.

Motivation: CORnet-S [21] was designed to model a primate ventral stream using a shallow ANN with four anatomically mapped areas (V1, V2, V4, IT). Each block used identical recurrent circuitry and skip connections and achieved state-of-the-art Brain-Score with only 15 layers of feedforward depth, demonstrating that within-area recurrence is the primary driver of biological alignment [21]. MouseNet [38] showed that the mouse visual system has a fundamentally different topology (shallow and parallel rather than deep and serial). VISp branches into five lateral areas at a similar hierarchical level before converging onto VISpor. CORM adopts a uniform 2-step recurrence across all HVA blocks and models only the cortical visual areas (VISp + grouped HVAs + VISpl + VISa + VISpor), receiving input directly from the existing dLGN module.

Input: 16 channels at 64×64 resolution from the dLGN module.

VISp Block Architecture: CORnet-S feedforward V1 preprocessor consisting of a 7×7 convolution (stride 2), 3×3 max-pooling (stride 2), and a 3×3 convolution. This matches VISp neuron counts of ~230K. Output: 64 channels at 16×16 resolution (no within-area recurrence).

Parallel HVA Blocks (5 recurrent blocks): Following VISp L5 output, five blocks process the input in parallel. Each implements the CORnet-S recurrent circuitry with exactly 2 recurrence steps. Channel counts are scaled proportionally to excitatory neuron counts from MouseNet using the formula $c_i = n_i / (l_{ix} \times l_{iy})$, where $l_{ix} \times l_{iy} = 256$ (the 16×16 feature map).

Functional Group	Excitatory Neurons (n)	Channels	n / Channel
Motion-biased	≈156,000	128	≈1,220
Pattern	≈42,000	48	≈875
Posterolateral	≈41,000	48	≈873
Bimodal spatiotemporal	≈26,000	32	≈838
Unassigned	≈25,000	32	≈781

Table 1: Channel allocation for parallel HVA blocks.

VISpor Block: Output is passed to the M2/MOs GRU and must therefore be a 64-d vector. We concatenate outputs from all HVA blocks (288 channels total) and apply the identical 2-step recurrent structure. A 1×1 projection reduces $288 \rightarrow 64$ channels. Final output: 64 channels at 8×8 resolution (global average pooling yields the 64-dimensional feature vector).

B Project Source Code

The source code for this project, including the Unity 3D visuo-motor arena Proof of Concept (PoC), the reinforcement learning training scripts, as well as additional exploratory experiments, is publicly available on GitHub.

The repository can be accessed at: https://github.com/Kriti1400/Syde552-Project/tree/main

C Individual Reports

Contribution	Time	Collaborators	Impact
dLGN Module Development	4 hrs	Entire group (architecture discussions)	Provided the first processing stage of the slow cortical pathway, grounding the module in biological constraints.
Visual Cortex Design & Appendix (CORnet-M)	8 hrs	—	Designed a novel mouse-specific visual cortex architecture (CORnet-M) inspired by CORnet-S; this became one of the key contributions of the project.
Motor Cortex (MOs/MOp) Integration	6 hrs	—	Designed motor cortex modules with persistent activity and dimensionality reduction, bridging perception and action.
Output Module (MLR + Spinal Circuits)	5 hrs	—	Designed thresholded gating logic for the MLR and finalized the motor signal output chain, completing the slow pathway.
Dual-Pathway Presentation Research	3 hrs	Entire group	Initial research into the 2-way motor pathway design formed the foundation for the group's preliminary project proposal.
Group Presentation	3 hrs	Entire group	Produced slides that communicated the dual-pathway architecture clearly to the class.
Pathway Integration & Writing	4 hrs	Olivia (discussed slow↔fast pathway interaction; wrote independently)	Consolidated the entire Slow Cortical Pathway section and integrated dual-pathway streams into the final group document.
Group Syncs	3 hrs	Entire group	Technical coordination meetings synchronized module architectures and project milestones across the team.
Literature Review	2 hrs	—	Grounded model parameters in biological data, ensuring biological plausibility (inspired by MouseNet's approach).
Peer Review & Editing	2 hrs	Entire group	Cross-module validation and structural refinement improved the cohesion of the team's final paper.
Experimentations (design POCs)	2 hrs	—	Validated model design decisions; not included in the final model but informed architectural choices.
Document Format Migration (Word to LaTeX)	3 hrs	—	Ported the entire document from .docx to ACM LaTeX format, set up build system, and organized all figures and sections.
Post-Migration Coordination & Refinement	—	Entire group	Coordinated with team members to integrate their minor revisions and changes into the ported LaTeX document.

Total personal time: 48 hours

Table 2: Individual Report - Vinayak Bector

Contribution	Time	Collaborators	Impact
Took primary ownership of designing the fast pathway architecture. Specified layer design representing biological architecture, including sSC, dSC, LP thalamus, CeA, vlPAG, PBG, BLA, dPAG, cuneiform, and implementing behaviour selection between freeze and escape.	~7 hours	Vinayak. Discussed how the fast and slow pathways should interact at a high level from a biological point of view. The pathways were fully individually designed and written. I designed the fast pathway. Kriti and Abirami. Discussed interfacing of the fast pathway with visual structures.	Established design for a core part of our visuomotor system, in particular for our application relating to threat/predation response.
Prototyped the fast pathway design, as well as output module.	~5 hours	Vinayak. Coordinated at a lower level what should computationally be passed between the fast and slow pathways. Prototyped code was individually written. I prototyped the code for the fast pathway.	Strengthened understanding of how biological structures should be represented computationally, and the biological validity of core architecture.
Resolved cross-pathway integration with slow pathway. Redefined parts of the model, including the separation of fast and slow pathways converging only at PAG/MLR after researching anatomical and experimental supporting evidence.	~3 hours	Vinayak raised questions regarding implementation, I researched the anatomy and proposed a solution.	Resolved structural ambiguity relating to both model and report, and ensured experimental literature grounded pathway separation decision.
Wrote the fast pathway of the motor report section covering the SC, freeze branch, escape branch, behavioral selection logic, and architecture summary diagram. Iterated on design and writing as the model developed.	~6 hours	None	Provided primary written and visual description of the fast pathway architecture and biological validity for the project in detail.
Contributed to initial research and ideation on project structure and focus, including application of predatory threat response, and possible data inputs including open-source datasets.	~3 hours	All members, group discussions and initial research.	Contributed to discussions outlining project's core framing in early stages, and what kinds of data inputs could be used.
Participated in group project meetings with Kriti, Abirami, and Vinayak to align on model architecture, section boundaries, and biological consistency across sections.	~3 hours	All members, group syncs.	Resolve cross-section inconsistencies to improve overall architecture and report coherence.
Engaged in peer review and editing of final report.	~3 hours	All members reviewed sections owned by others.	Identified overlapping sections to reduce redundancy, ensured the report flowed.

Total Time: ~30 hours

Table 3: Individual Report - Olivia Zheng

Contribution	Time	Collaborators (Indiv. Work Bolded)	Impact
Conducted preliminary research on potential ideas based on project requirements, course learnings, and external research.	~2 hours	All members. Everyone presented their initial ideas and **I presented my interest in sensorimotor integration with initial research**.	Came prepared with insights into brain systems/goals of interest to contribute effectively to the discussion and guide project decision-making forward.
Finalized project topic, prepared presentation notes on the motor system.	~3 hours	All members. Everyone split the topic into portions to focus on and **I presented research on the mouse motor system**.	In-depth understanding of the motor system dual pathways, with figures and links to papers to guide further project work.
Addressed professor feedback on project topic and split up work evenly within the team.	~2 hours	All members. Everyone aligned on project details and **I chose to specialize on the vision model implementation** with Abirami.	Further investigation into the motor system's recurrence feedback and how that connects to mouse defense mechanisms. Clear split of project work.
Conducted initial research on the vision model for mouse defense mechanisms, investigating existing datasets and models.	~5 hours	None.	Shared my findings with the team, specifically Sai, Hoang, and Abirami, during group discussions to sync on training input and vision architecture.
Coded initial prototyping POC of the vision model with biological grounding. Experimented with training parameters and generating synthetic training data.	~4 hours	Abirami. We worked independently to **investigate and implement our own prototypes**, in order to compare our findings and align on a final architecture.	Strengthened understanding of vision architecture like Difference-of-Gaussians kernels as a retinal filter, RGCs, SC superficial and deep components, PBGN and LPTN streams output.
Vision Research Discussions.	~1.5 hours	Abirami. **Discussed research and consolidated findings.**	Met with Abirami to align on initial research and POC findings, the overall architecture, and the computational model to discuss in the paper.
Conducted thorough research on input to the vision model and how mice observe looming vs sweeping stimuli, information processing, and connection to the mouse retina.	~4.5 hours	None. Aligned my findings with the rest of the team, particularly Sai, Abirami, and Hoang, for the visual input and architecture components.	Took primary ownership of the input to the vision model section, finding relevant research, data, and figures to support my findings, and aligning with the purpose of the report.
Conducted thorough research on the vision architecture connection to the motor systems and aligned with the team on handoff parts.	~4 hours	None. Aligned my findings with the rest of the team, particularly Abirami, Vinayak, and Olivia, for vision output and motor input.	Took primary ownership of the vision-motor model connection, finding relevant research and aligning with the report purpose.
Wrote half of the vision section of the report, integrating biological research with model details. Iterated on writing as the model developed.	~6 hours	None. Asked groupmates for review and feedback on related sections.	Contributed to the final submission, covering looming/sweeping input with biological grounding, and the architecture handoff to the motor system.
Cleaned up the code to reflect all updated parts of the architecture.	~2 hour	None.	Improved POC after alignment with team on updated architecture, like temporal processing over static images.
Actively participated in all group project meetings for alignment.	~4 hours	All members, **group syncs**.	Resolve inconsistencies to improve overall architecture and report coherence.
Engaged in peer review and editing of the final report.	~2 hours	All members **reviewed sections owned by others**.	Identified overlapping parts and ensured report flow across all sections.
Latex migration integration and refinement.	~2 hours	All members, **reviewed and formatted my own sections as well as the overall report**	Validated final report for minor errors in formatting, citations, image integration.

Total Time: ~42 hours

Table 4: Individual Report - Kriti Sodhi

Contribution	Time	Collaborators	Impact
Researched different ideas and topics for project brainstorming.	~2 hours	Everyone	Worked independently to research project ideas, evaluating pros and cons of different models. Discussed findings with the team, contributing to the decision to pursue vision and motor systems for predator detection.
Read & annotated core retinal biology sources.	~3.5 hours	Solo	Focused on Seabrook et al. (2017) and Huberman & Niell (2011), building an accurate understanding of retinal anatomy and RGC diversity to ground the biological foundation of the report.
Read & annotated vision computational modelling sources.	~4 hours	Solo	Reviewed 5+ papers on Difference of Gaussians receptive field modelling, the Naka–Rushton saturation model, and ON/OFF bipolar receptor types, connecting biological and computational literature. Research was critical to deriving the vision architecture.
Vision research discussions.	~1.5 hours	Kriti	Met with Kriti to align on biological vision findings, the overall architecture, and the computational model.
Synthesised research into visual system architecture.	~3 hours	Solo	Took ownership of creating the synthesised vision architecture diagram (Figure 4), providing a clear visual reference for the multi-stage processing pipeline that guided all subsequent writing for the vision section.
Wrote the Retinal Foundation section.	~4 hours	Solo	Wrote the Retinal Foundation section, detailing the mouse visual system's photoreceptor mosaic, RGC diversity, and retinotopic organization to establish the biological grounding for the downstream computational model.
Wrote the Vision Architecture section.	~4.5 hours	Solo	Led the vision architecture of the model, integrating retinal biology and computational theory, incorporating Weber-law contrast adaptation, the Naka–Rushton saturation function, Difference of Gaussians receptive field equations, and the parallel RGC channel framework that drives the model's survival behaviour outputs.
Vision architecture validation.	~2.5 hours	Solo	Cross-referenced each architectural choice: photoreceptor information, receptive field sizes, and RGC channel types against multiple sources to improve credibility.
Coded initial PyTorch prototype for modelling vision.	~4.5 hours	Kriti	Developed a proof-of-concept PyTorch implementation of the vision architecture, including DoG spatial filtering, RGC receptive fields, and ON/OFF pathway dynamics.
Updated code after improved architecture and team feedback.	~2 hours	Kriti	Debugged and revised the PyTorch prototype to more accurately reflect the updated architecture and incorporate feedback from the team.
Revising & editing report draft.	~2.5 hours	Everyone	Reviewed Kriti's vision introduction and motor connection sections, as well as contributions from other team members, to ensure consistency and clarity across the report.
Group syncs.	~4 hours	Everyone	Actively participated in group discussions on architecture decisions and project milestones to keep the team aligned throughout the project.
Presentation and research for vision.	~2.5 hours	Solo	Consolidated vision research and prepared content for the class presentation, covering the visual pathway and how the mouse vision system works.
Helped with LaTeX formatting.	~1.5 hours	Everyone	Formatted headings and individual report contribution to improve the final document flow.

Total Time: ~42 hours

Table 5: Individual Report - Abirami Karthikeyan

Contribution	Time	Collaborators	Impact
Preliminary Brainstorming & Research: Conducted initial research on potential ideas based on project requirements and external literature.	~2 hours	All members. Everyone presented their initial ideas and I presented my interest in coming up with an RL arena for the training plan.	Came prepared with insights into brain systems/goals of interest to contribute effectively to the discussion and guide project decision-making forward.
Topic Finalization & Presentation: Finalized the project topic and prepared presentation notes detailing the training methodology.	~3 hours	All members. Everyone split the topic into portions to focus on and I prepared a presentation on the training plan and idea of a unity arena.	Led to a better team understanding of Unity's capabilities and helped settle on a Deep RL training plan, resulting in a precursor to the final loss function.
Project Scope Alignment: Addressed professor feedback on the project topic and split up work evenly within the team.	~2 hours	All members. Everyone aligned on project details and I chose to focus on the training plan with Hoang.	Established a clear split of project work and became the team's primary owner for Unity prototyping and training environment development.
Biological Foundation Research: Researched the mouse defensive circuit, retino-collicular pathway, and PAG branches to establish a biological foundation.	~6 hours	Hoang. We synced to align on literature (e.g., Mobbs, Evans, Yilmaz, Campagner).	Established the biological foundation referenced throughout the report and ensured the training plan was biologically realistic.
Project Framing & Report Refinement: Collaborated on defining the project's core framing, explicitly tying model design choices to the intended goal. Proposed key revisions and added next-steps to the Future Work and Conclusion sections.	~3 hours	Hoang. Discussed the core framing before drafting the Purpose; helped refine after creation. Also proposed revisions and additions to future work and conclusion.	Made sure that the purpose was clear and our training plan adhered to it. Added several key points to the Future Work section to identify where we can go next with our model and prototypes.
Unity Set-up & Prototyping: Built and configured the 3D programmatic arena in Unity to establish the simulation environment.	~8 hours	Independently learned Unity to build the environment from the ground up. Hoang provided some insights and pointed out minor problems through the iterations.	Designed a biologically constrained environment that allows for continuous tracking of absolute coordinates. Ensured the feasibility of the training plan and evaluation metrics.
ML-Agents & Loss Function Design: Tested the Unity ML-Agents library to finalize the Deep RL objective function.	~6 hours	Discussion with Hoang on biologically grounding loss function. Independently learned and experimented with the Unity ML-Agents framework to test several proposed loss functions.	Formulated the mathematical constraints of the loss function. Justified training objectives by successfully balancing Foraging rewards against Death and Energy penalties to drive natural behavioural emergence.
Section Authorship (Training Plan): Wrote the bulk of the training plan, including the intro, RL Objective, and Evaluation Metrics.	~10 hours	Hoang helped review and provided some suggestions. I drafted through multiple revisions, making sure that each choice was backed up biologically.	Produced the core training methodology. Defined the physical constraints of the simulation (Intro), formulated the mathematical balance of neuroecological survival pressures (RL Objective), and established the *in silico* lesion testing framework (Evaluation Metrics).
Unity Sync with Hoang	~2 hours	Getting connected on details with Unity and how we can biologically ground it. Discussion on loss function and data inputs.	Maintained tight integration between the training methodology and the biological architecture.
Group meetings	~4 hours	All members. Verified that training using Unity was the most viable approach.	Resolved cross-section inconsistencies and ensured biological framing stayed consistent throughout the report.
Engaged in peer review, LaTeX migration, and editing of the final report.	~2 hours	All members reviewed sections owned by others.	Identified overlapping parts and ensured report flow across all sections.
Total Time	**~48 hours**		

Table 6: Individual Report - Saivenkat Jilla

Contribution	Time	Collaborators	Impact
Conducted a full review of the project scope, rubric, and teammates' drafted sections to assess progress and identify outstanding contributions needed.	~4 hours	None.	Identified that Background, Purpose, Future works, Conclusion and Training Plan needed the most work and allocated effort accordingly, also acknowledge other's part idea so I can make the project connected throughout.
Researched the mouse defensive circuit including the retino-collicular pathway, PAG branches, and goal-driven modelling literature to prepare for writing.	~9 hours	Sai. We split reading by topic and compared findings before writing.	Established the biological foundation referenced throughout the report.
Wrote and revised the Background section across multiple drafts with APA citations and reference list.	~5 hours	Written Independently. Bector reviewed.	Produced the final Background section supporting the Biological Grounding rubric criterion.
Wrote and revised the Purpose section, balancing specificity about the model without redundantly repeating the Background.	~3 hours	Sai. We discussed framing together before I drafted independently.	Tied all design choices to a clear stated goal, addressing the Clarity of Purpose rubric criterion.
Contributed initially to the visual network architecture section before role was reassigned to Training Plan to balance group workload.	~3 hours	None.	Early visual architecture research directly informed how training inputs and stimulus classes were framed. Leave a rough work for later people.
Supported the Training Plan section by providing biological grounding research, reviewing drafts, and contributing suggestions on the loss function framing and stimulus class design. Sai led the writing and Unity implementation.Refined and edited the full section for clarity and biological accuracy.	~5 hours	Sai led. I reviewed drafts and synced on biological accuracy of the loss function and training inputs.	Produced the core training methodology addressing the Feasibility of Training Plan rubric criterion. Refinements ensured the loss function, evolutionary justification, and Unity environment rationale were clearly connected to the biological architecture across the full report.
Wrote the Conclusion section independently, revised multiple times for tone and coherence.	~3 hours	Written independently. Sai, Bector, and Kriti reviewed and gave feedback, suggestion.	Provided an impactful closing to the report framing the project as a testable biological hypothesis.
Wrote the Future Work section independently, identifying key limitations and research directions.	~2 hours	Written independently. Bector and Sai reviewed.	Strengthened the report by grounding limitations in specific biological constraints missed by the current design.
Actively participated in group meetings for alignment on architecture decisions and report structure. Reviewed other parts, added figures, citations refinement.	~9 hours	All members.	Resolved cross-section inconsistencies and ensured biological framing stayed consistent throughout. Refine the report in general. Fixed coherence throughout the report.

Total Time: ~43 hours

Table 7: Individual Report - Hoang Duy Le